

Wayland Kiosk Setup Guide for Raspberry Pi Zero 2 W (DietPi)

This guide provides a robust, hardware-accelerated portrait kiosk configuration. It resolves icon rendering issues and permanently eliminates the mouse cursor via kernel-level masking.

System Requirements

- **Hardware:** Raspberry Pi Zero 2 W + Display (Portrait).
- **OS:** DietPi (Debian Bookworm).
- **Target:** An accessible web dashboard URL.

Step 1: Install Dependencies

```
sudo apt update
sudo apt install -y cage cog wlr-randr fonts-noto-color-emoji fontconfig
```

Note: `fonts-noto-color-emoji` is critical to prevent icon glyphs from appearing as "tofu" (missing squares) in React/web dashboards.

Step 2: Kernel-Level Cursor Removal

To stop Cage from spawning a cursor, we tell the kernel to hide pointer capabilities for all devices:

```
sudo nano /etc/udev/rules.d/99-hide-cursor.rules
```

Add these lines:

```
ENV{ID_INPUT_POINTINGSTICK}="0"
ENV{ID_INPUT_MOUSE}="0"
```

Apply the change:

```
sudo udevadm control --reload-rules && sudo udevadm trigger
```



Step 3: Startup Script

Create `/home/dietpi/start-kiosk.sh` :

```
#!/bin/bash
sleep 1
wlr-randr --output HDMI-A-1 --transform 90

# Bypass security for local dashboard assets
export WEBKIT_CHOICE_MIXED_CONTENT_POLICY=0
export WEBKIT_DISABLE_WEB_SECURITY=1

exec /usr/bin/cog "http://YOUR_DASHBOARD_URL_HERE"
```

Make it executable:

```
chmod +x /home/dietpi/start-kiosk.sh
```



Step 4: Systemd Service

```
sudo nano /etc/systemd/system/kiosk.service
```

```
[Unit]
Description=Wayland Cage Cog Kiosk Service
After=systemd-user-sessions.service network.target systemd-logind.service
Requires=systemd-user-sessions.service

[Service]
User=root
Type=simple
Environment=XDG_RUNTIME_DIR=/run/user/0
Environment=LIBSEAT_BACKEND=builtin
Environment=COG_PLATFORM_FDO_VIEW_FULLSCREEN=1
ExecStartPre=/bin/mkdir -p /run/user/0
ExecStartPre=/bin/chmod 700 /run/user/0
ExecStart=/usr/bin/cage -d -s -- /home/dietpi/start-kiosk.sh
Restart=always
```

```
[Install]
```

```
WantedBy=multi-user.target
```



Step 5: Final Activation

```
sudo fc-cache -fv
```

```
sudo systemctl daemon-reload
```

```
sudo systemctl enable --now kiosk.service
```